Interactive Book Data Visualization Dashboard

Project Report

Dataset: books_data.csv
1,000 rows | 8 columns
Generated on 06 November 2026

Python | Pandas | NumPy | Matplotlib | Seaborn | Plotly | Streamlit

1. Introduction

This report accompanies an interactive dashboard that turns a CSV file of book data into charts, summary statistics and downloadable results. The dashboard uses only the real data in the file: it never creates sample data and never fills in missing values. Rows that lack a value needed for a chart are skipped for that chart only.

Main features: bar, line, pie, histogram, scatter and box charts; dataset summary and statistical insights; interactive sidebar filters; a CSV upload option; and downloads for charts (PNG / HTML), tables (CSV), a Markdown analysis report, a PDF report and a ZIP bundle.

2. Dataset Overview

Measure	Value
Rows	1,000
Columns	8
Numeric columns	3
Text columns	4
Missing cells	0
Duplicate rows	0
Memory (KB)	192.5

Column summary

Column	Type	Non-null	Missing	Missing %	Unique
title	str	1000	0	0	999
price_gbp	float64	1000	0	0	903
rating_stars	int64	1000	0	0	5
rating_text	str	1000	0	0	5
availability	str	1000	0	0	1
in_stock	bool	1000	0	0	1
page_number	int64	1000	0	0	50
product_url	str	1000	0	0	1000

3. Statistical Summary

Numeric columns

Column	Count	Mean	Std	Min	25%	Median	75%	Max	Skew
price_gbp	1000	35.07	14.45	10	22.11	35.98	47.46	59.99	-0.04
rating_stars	1000	2.92	1.44	1	2	3	4	5	0.07
page_number	1000	25.50	14.44	1	13	25.50	38	50	0

Text and true/false columns

Column	Unique	Most common	Count	Share %
title	999	The Star-Touched Queen	2	0.20
rating_text	5	One	226	22.60
availability	1	In stock	1000	100
in_stock	1	True	1000	100
product_url	1000	https://books.toscrape.com/catalogue/a-light-in-the-att	1	0.10

4. Visualizations

Each chart below is drawn from the data with Matplotlib and Seaborn. The dashboard shows interactive Plotly versions of the same charts.

4.1 Bar chart

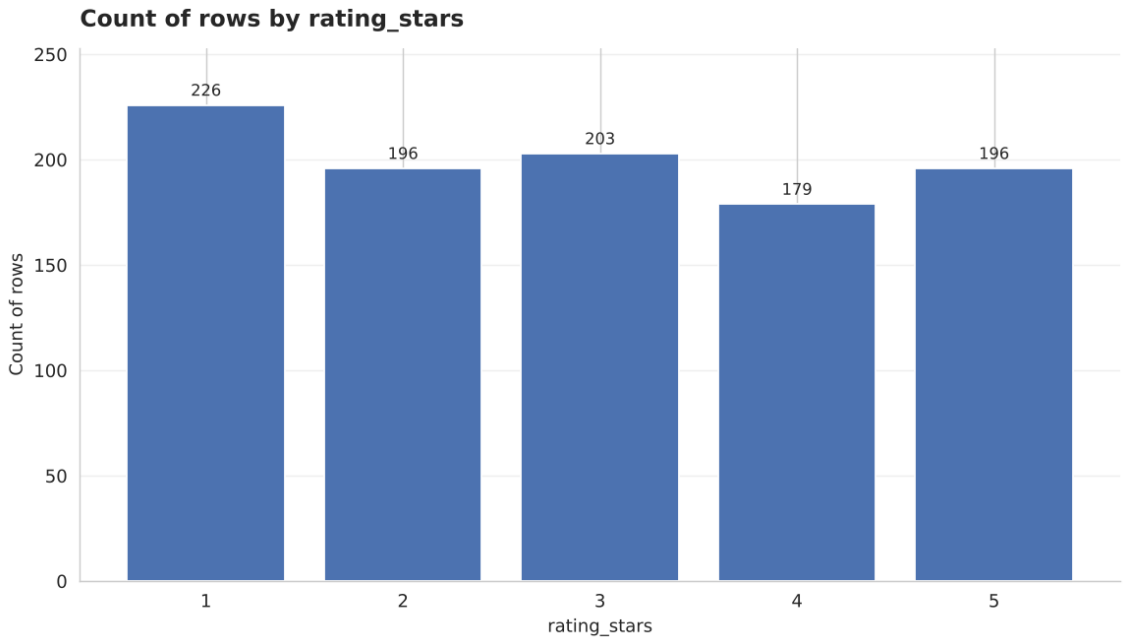


Figure 1. Highest: rating_stars = 1 (226). Lowest: rating_stars = 4 (179).

4.2 Line chart



Figure 2. Highest point: 49 (41.27). Lowest point: 31 (26.66). 50 distinct X values.

4.3 Pie chart

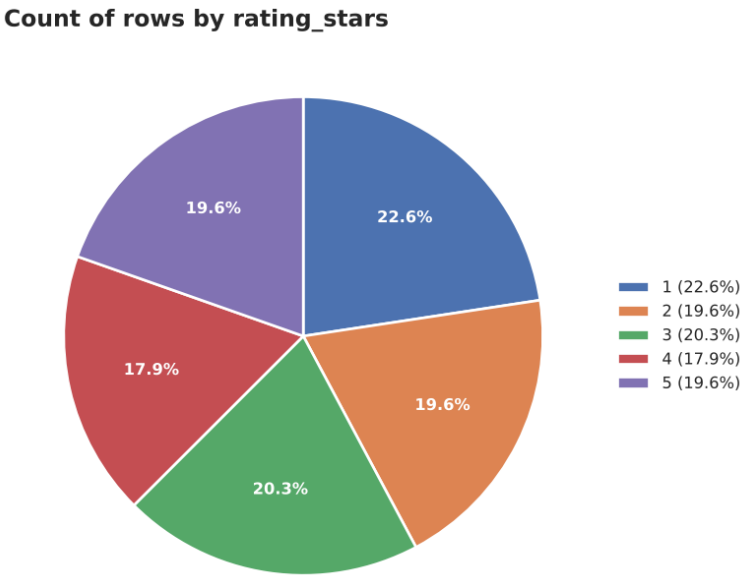


Figure 3. Largest slice: rating_stars = 1 (22.6% of the total). 5 slices in total.

4.4 Histogram



Figure 4. 1,000 values; mean 35.07, median 35.98, range 10 to 59.99. Most common range: 36.66 to 38.33 (45 values).

4.5 Scatter plot

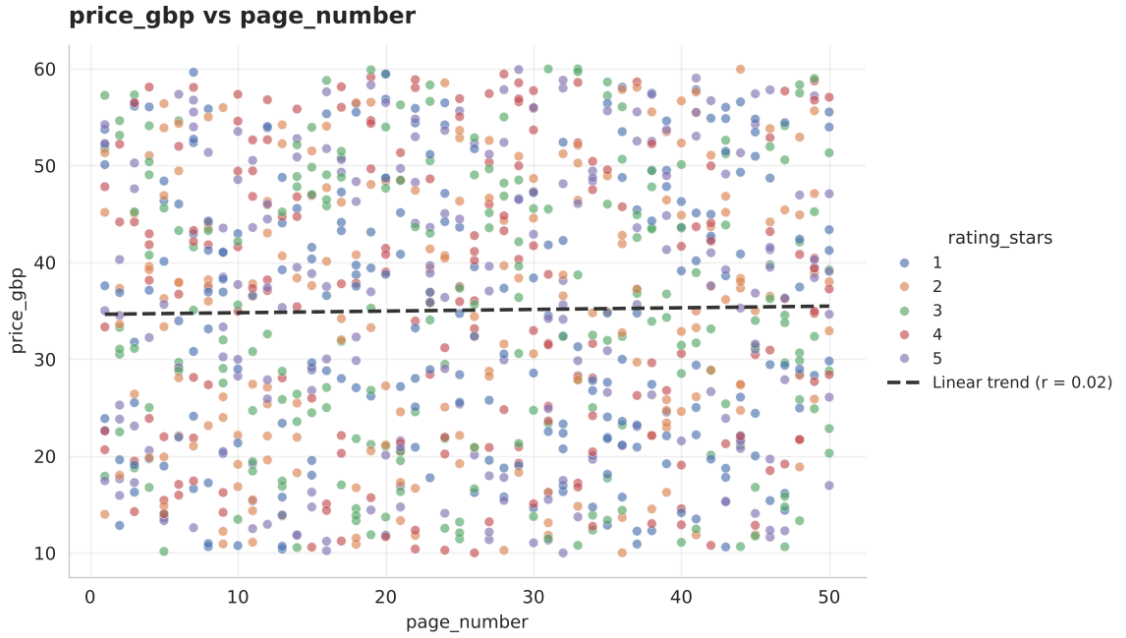


Figure 5. 1,000 points plotted. Pearson r = 0.02 (negligible positive linear relationship). On average, y changes by 0.017 for each +1 in x. This shows association, not cause and effect.

4.6 Box plot

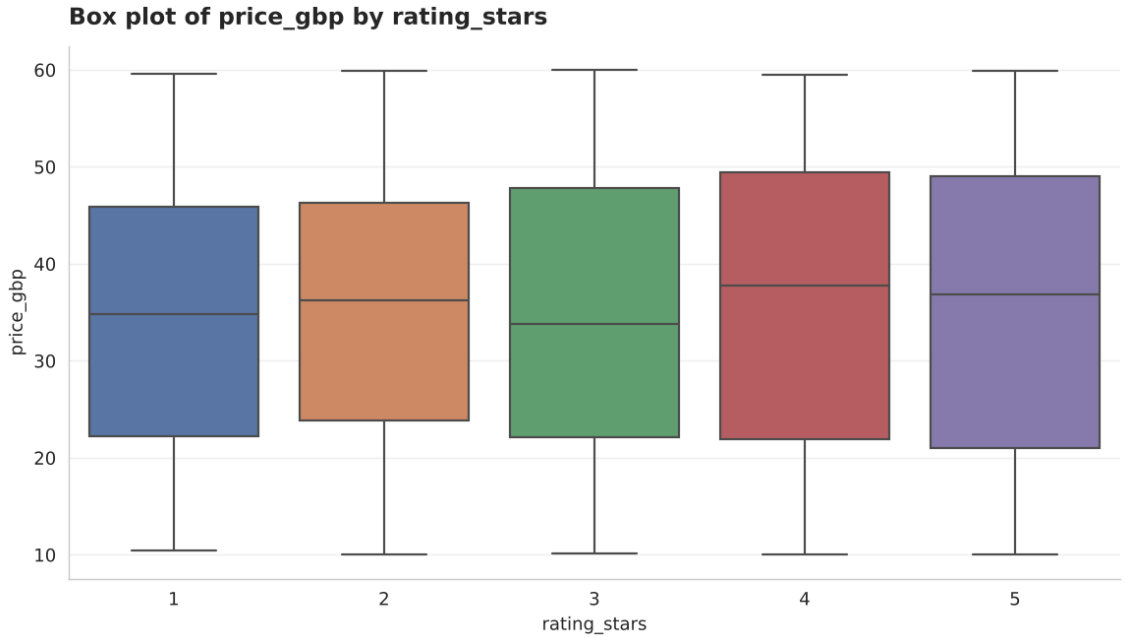


Figure 6. Highest median: rating_stars = 4 (37.80). Lowest median: rating_stars = 3 (33.78). 0 values fall outside the overall 1.5xIQR fences.

4.7 Correlation heatmap

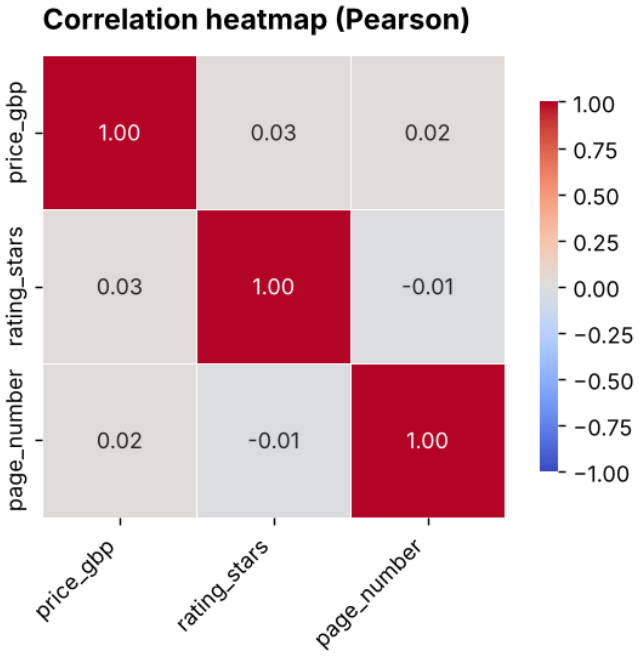


Figure 7. Pearson correlation between numeric columns that vary.

5. Key Insights

Every statement below was calculated from the data when this report was generated.

- The data has 1,000 rows and 8 columns (3 numeric, 4 text, 1 true/false).
- There are no missing values in any column.
- There are no exact duplicate rows.

- Columns that are almost unique but contain repeats: 'title' (1 repeated value(s)).
- Columns with only one distinct value (no variation, so not useful for comparisons): availability, in_stock.
- price_gbp: mean 35.07, median 35.98, range 10 to 59.99, standard deviation 14.45; the distribution is roughly symmetric; no outliers by the 1.5xIQR rule.
- rating_stars: mean 2.92, median 3, range 1 to 5, standard deviation 1.43; the distribution is roughly symmetric; no outliers by the 1.5xIQR rule.
- page_number: mean 25.50, median 25.50, range 1 to 50, standard deviation 14.44; the distribution is roughly symmetric; no outliers by the 1.5xIQR rule.
- rating_stars: 5 distinct values; the most common is '1' (226 rows, 22.6% of non-missing).
- rating_text: 5 distinct values; the most common is 'One' (226 rows, 22.6% of non-missing).
- Strongest linear relationship between numeric columns: price_gbp and rating_stars (Pearson r = 0.03, negligible). Correlation shows association only - not cause and effect.
- Average price_gbp by rating_stars: highest for rating_stars = 4 (36.09), lowest for rating_stars = 1 (34.56); gap = 1.53 (groups with fewer than 5 rows are ignored).

6. Methodology and Tools

- Pandas reads the CSV and builds all summaries; NumPy supports the calculations.
- Plotly draws the interactive charts; Matplotlib and Seaborn draw the downloadable images.
- Streamlit provides the dashboard, filters and download buttons.
- Outliers use the 1.5 x IQR rule. Correlation uses Pearson's r (Spearman is available in the app).
- Category labels that are missing are shown as '(missing)'; numeric gaps are left empty, never filled.

7. Limitations and Notes

- Results describe only the rows in the file (or the rows left after filtering); they do not necessarily describe all books everywhere.
- Correlations show association, not cause and effect.
- If the file has a column such as page_number, it is the position in the source listing, not a date, so line charts over it show order in the listing, not time.
- Columns where every value is the same show no variation and cannot be compared.